

**NABEELA MEERAL M 2024-AIDS** ▾**N2****Started on** Friday, 22 August 2025, 10:12 AM**State** Finished**Completed on** Friday, 22 August 2025, 10:16 AM**Time taken** 3 mins 57 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  void function (int n)
3  {
4      int c=0;
5      int i= 1;
6      c++;
7      int s =1;
8      c++;
9      while(s <= n)
10     {
11         c++;
12         i++;
13         c++;
14         s += i;
15         c++;
16     }
17     c++;
18     printf("%d",c);
19 }
20 int main()
21 {
22     int n;
23     scanf("%d",&n);
24     function(n);
25     return 0;
26 }
27
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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NABEELA MEERAL M 2024-AIDS ▾

N2**Started on** Friday, 22 August 2025, 10:54 AM**State** Finished**Completed on** Friday, 22 August 2025, 11:00 AM**Time taken** 5 mins 4 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  void func(int n)
3  {
4      int c=0;
5      c++;
6      if(n==1)
7      {
8          c++;
9      }
10     else
11     {
12         for(int i=1; i<=n; i++)
13         {
14             for(int j=1; j<=n; j++)
15             {
16                 c++;
17                 c++;
18                 c++;
19                 c++;
20                 break;
21             }
22             c++;
23         }
24         c++;
25     }
26     printf("%d",c);
27 }
28 int main()
29 {
30     int n;
31     scanf("%d",&n);
32     func(n);
33     return 0;
34 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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NABEELA MEERAL M 2024-AIDS ▾

N2**Started on** Friday, 22 August 2025, 10:50 AM**State** Finished**Completed on** Friday, 22 August 2025, 10:54 AM**Time taken** 3 mins 52 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  void Factor(int num)
3  {
4      int c=0;
5      for (int i = 1; i <= num; ++i)
6      {
7          c++;
8          c++;
9          if (num % i == 0)
10         {
11             c++;
12         }
13     }
14     c++;
15     printf("%d", c);
16 }
17 int main()
18 {
19     int num;
20     scanf("%d", &num);
21     Factor(num);
22     return 0;
23 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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NABEELA MEERAL M 2024-AIDS ▾

N2**Started on** Friday, 22 August 2025, 10:27 AM**State** Finished**Completed on** Friday, 22 August 2025, 10:45 AM**Time taken** 17 mins 57 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2  void function(int n)
3  {
4      int counter=0;
5      int c = 0;
6      counter++;
7      for (int i = n / 2; i < n; i++)
8      {
9          counter++;
10         for (int j = 1; j < n; j = 2 * j)
11         {
12             counter++;
13             for (int k = 1; k < n; k = k * 2)
14             {
15                 counter++;
16                 c++;
17                 counter++;
18             }
19             counter++;
20         }
21         counter++;
22     }
23     counter++;
24     printf("%d",counter);
25 }
26 int main()
27 {
28     int n;
29     scanf("%d", &n);
30     function(n);
31     return 0;
32 }
33
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[NABEELA MEERAL M 2024-AIDS](#) ▾**N2****Started on** Friday, 22 August 2025, 10:45 AM**State** Finished**Completed on** Friday, 22 August 2025, 10:49 AM**Time taken** 3 mins 53 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  void reverse(int n)
3  {
4      int c=0;
5      int rev = 0;
6      c++;
7      int remainder;
8      while (n != 0)
9      {
10         c++;
11         remainder = n % 10;
12         c++;
13         rev = rev * 10 + remainder;
14         c++;
15         n/= 10;
16         c++;
17     }
18     c++;
19 c++;
20 printf("%d",c);
21 }
22 int main()
23 {
24     int n;
25     scanf("%d",&n);
26     reverse(n);
27     return 0;
28 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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